



Concept-Guided Medical Image Caption Generation Using UMLS Concept Detection and Image-Concept Fusion

Senior 2 Graduation Project

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled "Concept-Guided Medical Image Caption Generation Using UMLS Concept Detection and Image-Concept Fusion", prepared by "Bashar Alsaleh", was carried out under my supervision at the Faculty of Artificial Intelligence Engineering, Syrian Private University, in partial fulfillment of the requirements for the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science.

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Acknowledgements

Dedication

Dedication

الملخص

مع ImageCLEFmedical Caption 2026 يعالج هذا المشروع مهمة توليد الوصف الطبي للصور الشعاعية ضمن إطار كتمثيل دلالي وسيط بين الصورة الطبية والنص الناتج. تتبع أهمية هذه المسألة من أن UMLS التركيز على الاستفادة من مفاهيم الوصف الطبي الآلي لا يقتصر على إنتاج جملة لغوية سليمة، بل يتطلب الحفاظ على المعنى السريري، وتجنب إغفال المفاهيم التشريحية أو المرضية المهمة، والحد من توليد معلومات طبية غير مدعومة بالصورة. لذلك يتعامل البحث مع التوليد الطبي باعتباره مسألة رؤية ولغة تحتاج إلى ربط منظم بين الدليل البصري والمفاهيم الطبية القياسية.

فقط، ثم تنتقل إلى تدريب النماذج على مفاهيم UMLS يقترح العمل منهجية تدريجية تبدأ من نماذج نصية تعتمد على مفاهيم متوقعة من كاشف المفاهيم المطور في مشروع التخرج الأول، وصولاً إلى نموذج دمج نهائي يجمع بين ميزات الصورة تم تصميم التجارب بحيث تميز بين T5 المقروءة نصياً داخل نموذج UMLS ومفاهيم BiomedCLIP المستخرجة باستخدام incomplete أو noisy والحالة العملية التي تستخدم مفاهيم متوقعة، Ground-Truth التي تستخدم مفاهيم Oracle حالة هذا التصميم بتحليل أثر جودة المفاهيم على جودة الوصف، وقياس الفجوة بين الحد الأعلى النظري والنظام القابل للتطبيق.

الخام يحسن جودة التوليد بوضوح. كما أن التدريب على CUI بدلاً من رموز UMLS أظهرت النتائج أن إدخال أسماء مفاهيم Exp6 المفاهيم المتوقعة أو الدمج بين مفاهيم صحيحة ومتوقعة يزيد من متانة النموذج أمام الضجيج. حقق نموذج الدمج النهائي مقدار ROUGE-L 0.3205 أفضل نتيجة شاملة على مجموعة التحقق الداخلية، حيث بلغ GT باستخدام الصورة ومفاهيم مقدار 0.5850. أما Dictionary-based UMLS-F1 مقدار 0.4197 و BLEURT مقدار 0.6656 و BERTScore-F1 و منافساً وحقق توازناً جيداً بين التشابه الدلالي واتساق المفاهيم Exp6 في الحالة العملية باستخدام المفاهيم المتوقعة، فقد بقي نموذج يمثل GT أن إعداد ImageCLEFmedical Caption 2026 الطبية. وتؤكد مقارنة النتائج مع الأنظمة المرجعية ونتائج لوحة حداً أعلى رقمياً، بينما تمثل الإعدادات المعتمدة على المفاهيم المتوقعة النظام العملي القابل للمقارنة.

خلص تحليل الأخطاء إلى أن المفاهيم المفقودة أكثر ضرراً من المفاهيم الزائدة في كثير من الحالات، إذ يؤدي غياب المفاهيم التشريحية أو المرضية الأساسية إلى أوصاف غير دقيقة أو إلى هلوسة في نوع الصورة أو منطقة الجسم. في المقابل، ساعدت الميزات البصرية في نموذج الدمج على تخفيف أثر ضجيج المفاهيم المتوقعة وتحسين الاتساق الطبي مقارنة بالنماذج النصية فقط. وجودة توليد الأوصاف UMLS وبذلك يقدم هذا العمل مساهمة عملية وأكاديمية في دراسة العلاقة بين جودة اكتشاف مفاهيم الطبية.

دمج الصورة والمفاهيم، تحليل الأخطاء، BiomedCLIP، T5، ImageCLEFmedical Caption 2026، UMLS الكلمات المفتاحية: توليد الوصف الطبي، مفاهيم

Abstract

This thesis addresses medical image caption generation in the context of ImageCLEFmedical Caption 2026, with a specific focus on using UMLS concepts as structured semantic guidance between radiology images and generated captions. The problem is important because medical caption generation is not only a language-generation task; it also requires the model to preserve clinically relevant concepts, avoid omitting important anatomical or pathological information, and reduce unsupported medical hallucination. For this reason, the thesis treats medical caption generation as a vision-language problem controlled by intermediate medical concepts rather than as direct image-to-text generation only.

The proposed methodology follows a staged experimental design. It begins with text-only generation from raw CUI identifiers, then introduces readable UMLS concept names, then fine-tunes caption generators on predicted concepts produced by the Senior Project I concept detector, and finally develops an image-concept fusion model. The final model combines frozen BiomedCLIP visual features, a learned visual prefix adapter, and T5-based concept-guided caption generation. The experiments explicitly distinguish between oracle settings that use ground-truth concepts and practical settings that use predicted noisy or incomplete concepts. This design makes it possible to quantify the effect of concept quality on caption quality and to separate the upper-bound value of accurate concepts from the deployable predicted-concept scenario.

The results show that converting raw CUIs into readable UMLS terms substantially improves caption generation. Robust predicted-concept fine-tuning and mixed ground-truth/predicted training further improve performance under noisy concept guidance. The strongest overall result was achieved by Exp6 using image features together with ground-truth UMLS concepts, reaching ROUGE-L = 0.3205, BERTScore-F1 = 0.6656, BLEURT = 0.4197, and dictionary-based UMLS-F1 = 0.5850 on the internal validation subset. In practical predicted-concept settings, Exp6 remained competitive and provided the strongest balance between lexical overlap, semantic similarity, and medical concept consistency among the deployable configurations. Comparisons with selected 2024/2025 reference systems and the official ImageCLEFmedical Caption 2026 leaderboard show that the ground-truth concept setting should be interpreted as an oracle upper-bound, whereas predicted-concept fusion represents the practical configuration.

The error analysis indicates that missing concepts are often more harmful than additional noisy concepts, especially when the missing information corresponds to anatomy, modality, or clinically important findings. Visual information helps reduce the negative effect of predicted-concept errors, while concept guidance improves medical grounding compared with image-only generation. Overall, the thesis demonstrates that concept-guided image-caption fusion is a promising and

interpretable direction for medical caption generation, particularly when the relationship between concept detection quality and downstream language generation is analyzed explicitly.

Keywords: medical image captioning, UMLS concepts, ImageCLEFmedical Caption 2026, T5, BiomedCLIP, image-concept fusion, error analysis.

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List of Symbols

Symbol / Term	English Equivalent	Meaning / Description
AI	Artificial Intelligence	Computational methods that enable machines to perform tasks requiring intelligence.
ML	Machine Learning	A field of AI focused on learning patterns from data.
DL	Deep Learning	Machine learning based on multi-layer neural networks.
NLP	Natural Language Processing	Computational processing and generation of human language.
VLM	Vision-Language Model	Model that jointly processes visual and textual information.
UMLS	Unified Medical Language System	Controlled biomedical terminology system used to represent medical concepts.
CUI	Concept Unique Identifier	Unique identifier assigned to a UMLS concept.
ROCOv2	Radiology Objects in Context version 2	Radiology image-caption dataset used in ImageCLEFmedical Caption.
PMC	PubMed Central	Open-access biomedical article source used in the dataset background.
CNN	Convolutional Neural Network	Image model based on convolutional feature extraction.
DenseNet	Densely Connected Convolutional Network	CNN backbone used as a baseline in related medical imaging tasks.
ViT	Vision Transformer	Transformer architecture that processes images as patch sequences.
Swin	Shifted Window Transformer	Hierarchical vision transformer using shifted local attention windows.
BiomedCLIP	Biomedical Contrastive Language-Image Pretraining	Biomedical vision-language encoder used to extract image features.
CLIP	Contrastive Language-Image Pretraining	Image-text pretraining framework based on contrastive learning.
T5	Text-to-Text Transfer Transformer	Encoder-decoder transformer used for caption generation.
GT	Ground Truth	Manually provided target annotation or reference concept set.
Oracle	Oracle Upper-bound	Evaluation setting that uses ground-truth information not available at deployment.
Pred	Predicted	Concepts generated by a trained concept detection model.
Micro	Micro-optimized concept set	Predicted concept setting optimized for micro-level accuracy.
Coverage	Coverage-oriented concept set	Predicted concept setting emphasizing broader concept recall.
CUI-to-term mapping	Concept name mapping	Conversion of UMLS CUIs into readable medical terms.
Prompt	Input prompt	Textual input passed to the caption generator.
Visual prefix	Learned visual token prefix	Set of projected image tokens prepended to the T5 encoder input.
Adapter	Projection adapter	Trainable module that maps image embeddings into language-model space.
ROUGE-1	Recall-Oriented Understudy for Gisting Evaluation-1	Unigram overlap metric between generated and reference captions.
ROUGE-2	ROUGE bigram overlap	Bigram overlap metric between generated and reference captions.

Symbol / Term	English Equivalent	Meaning / Description
ROUGE-L	ROUGE longest common subsequence	Metric based on longest common subsequence overlap.
BLEU	Bilingual Evaluation Understudy	Precision-oriented n-gram metric for generated text.
BERTScore	BERT-based semantic text similarity	Embedding-based metric for semantic similarity.
BLEURT	BERT-based learned evaluation metric	Learned text-generation evaluation metric trained from human judgments.
MedCAT	Medical Concept Annotation Toolkit	Tool used in the official challenge to extract UMLS entities for factuality scoring.
UMLS-F1	UMLS Concept F1	Concept overlap score between reference and generated captions.
Dict-UMLS F1	Dictionary-based UMLS F1	Internal UMLS consistency proxy based on dictionary matching.
AlignScore	Factual consistency alignment score	Metric for evaluating factual consistency between text units.
Similarity	Image-caption similarity	Official metric based on visual-text embedding similarity.
TP	True Positive	Correctly predicted positive concept or entity.
FP	False Positive	Incorrectly predicted positive concept or entity.
FN	False Negative	Missed concept or entity.
Precision	Positive predictive value	Fraction of predicted positives that are correct.
Recall	Sensitivity	Fraction of actual positives recovered by the model.
F1	Harmonic mean of precision and recall	Balanced metric combining precision and recall.
ASL	Asymmetric Loss	Loss designed for imbalanced multi-label classification.
EMA	Exponential Moving Average	Weight averaging strategy for stable training.
bf16	Brain floating point 16-bit	Mixed precision format used on modern accelerators.
Validation subset	Internal validation sample	Subset of 3,000 validation samples used for controlled evaluation.
Hallucination	Unsupported generated content	Generated medical information not supported by the image/reference concepts.
Missed concept	Omitted reference concept	Relevant concept present in the reference but absent from generated output.
Long-tail	Long-tailed label distribution	Distribution with many rare labels and few frequent labels.

Chapter 1: Introduction

This chapter introduces the general context of the research and explains the scientific problem addressed in this thesis. It motivates concept-guided medical image caption generation, defines the research objectives and contributions, and presents the organization of the remaining chapters.

1.1 Background and Motivation

Artificial intelligence has become a major component of modern medical image analysis. Deep learning models can assist in classification, retrieval, report generation, triage, and decision-support research. Among these tasks, medical image caption generation is particularly challenging because it requires the system to convert visual information into concise clinically meaningful language.

Radiology images are different from natural images. They are often grayscale, modality-dependent, and difficult to interpret without clinical background knowledge. A captioning system must therefore understand anatomical regions, imaging modalities, visual findings, and sometimes the relationship between devices, lesions, and surrounding tissue. A fluent caption that misses an important finding is not sufficient in a medical setting.

The ImageCLEFmedical Caption task provides a useful research framework for this problem because it connects radiology images with captions and UMLS concept annotations. These concepts can act as a bridge between visual information and medical language. A model can use concepts as structured guidance rather than relying only on raw image pixels.

This thesis continues the work developed in Senior Project I. The previous project focused on medical concept detection and produced predicted UMLS concepts using a frequency-aware Swin Transformer model. In Senior Project II, those concepts become the input to a caption generation pipeline. The central idea is to study how concept quality affects downstream caption generation.

This aspect is important for the present project because concept-guided medical caption generation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, concept-guided medical caption generation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

1.2 Scientific Problem of the Research

The scientific problem addressed in this thesis is how to generate accurate, concise, and medically grounded captions for radiology images when the semantic guidance may be correct, incomplete, or noisy. In an ideal setting, the model receives ground-truth UMLS concepts. In a practical setting, however, these concepts must be predicted automatically by a concept detector.

The problem is therefore not limited to building a caption generator. It also includes studying the dependency between concept detection quality and caption generation quality. If predicted concepts omit a relevant anatomical term or add an unrelated finding, the caption generator may produce an incomplete or hallucinated description.

A second difficulty is that automatic text metrics measure different aspects of quality. ROUGE and BLEU focus on lexical overlap; BERTScore and BLEURT measure semantic similarity; UMLS-based metrics attempt to measure medical concept consistency. A strong system should be interpreted across all these complementary metrics rather than through a single score.

This aspect is important for the present project because the scientific problem affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, the scientific problem must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

1.3 Research Objectives

- To build a controlled experimental pipeline for UMLS concept-guided medical image caption generation.
- To compare raw CUI inputs with readable UMLS term inputs and measure the value of semantic term enrichment.
- To analyze the gap between oracle ground-truth concept guidance and practical predicted-concept guidance.
- To investigate whether training on predicted concepts improves robustness against noisy concept inputs.
- To integrate visual information through BiomedCLIP features and a visual prefix adapter.
- To evaluate the final models using lexical, semantic, and medical concept-consistency metrics.

- To compare the results with previous ImageCLEFmedical reference systems and the official 2026 leaderboard in a careful and scientifically valid manner.

1.4 Research Questions

- How much does converting UMLS CUIs into readable UMLS terms improve caption generation?
- How large is the performance gap between ground-truth concept guidance and predicted-concept guidance?
- Can predicted-concept fine-tuning or mixed training improve robustness to noisy concept inputs?
- Does adding image information through BiomedCLIP visual features improve practical caption generation compared with concept-only models?
- Which types of errors are most damaging: missing concepts, noisy concepts, missed generated UMLS entities, or hallucinated generated entities?

1.5 Research Contributions

- A complete staged experimental study from raw CUI-only generation to image-concept fusion.
- A direct comparison between oracle ground-truth concept guidance and practical predicted-concept guidance.
- A robust mixed-training strategy that combines ground-truth and predicted concept prompts.
- An image-concept fusion architecture using BiomedCLIP features projected into T5 encoder space as visual prefix tokens.
- A detailed error analysis connecting concept detection errors to caption generation quality.
- A final comparison with selected 2024/2025 references and the official ImageCLEFmedical Caption 2026 leaderboard.

This aspect is important for the present project because research contributions affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, research contributions must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

1.6 Beneficiaries and Scope

The direct beneficiaries of this research include students and researchers working on medical image captioning, medical vision-language modeling, and concept-based image understanding. The work is also useful for future systems that use structured medical concepts as intermediate representations for captioning, retrieval, indexing, or report support.

The scope of the thesis remains academic and experimental. The generated captions are not intended for clinical deployment or independent diagnosis. The system is evaluated on an internal validation subset and compared with official leaderboard systems only for contextual analysis.

This aspect is important for the present project because scope and beneficiaries affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, scope and beneficiaries must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

1.7 Organization of the Thesis

Chapter 2 reviews the theoretical background. Chapter 3 discusses related work and the research gap. Chapter 4 presents the methodology and experimental design. Chapter 5 reports the quantitative results and comparisons. Chapter 6 provides error analysis and discussion. Chapter 7 concludes the thesis and presents future work.

1.8 Chapter Summary

This chapter introduced the motivation, scientific problem, objectives, and contributions of the thesis. It emphasized that the core research question is not only whether a caption can be generated, but how concept quality affects the medical correctness and semantic grounding of that caption.

Chapter 2: Theoretical Background

This chapter reviews the foundations needed to understand the proposed approach, including medical image captioning, UMLS concepts, transformer-based generation, biomedical vision-language encoders, visual prefix learning, concept noise, and evaluation metrics.

2.1 Medical Image Caption Generation

Medical image caption generation aims to produce a short natural-language description for a medical image. Unlike general image captioning, the generated sentence must be clinically meaningful and should avoid unsupported findings. The task requires both visual recognition and domain-specific language generation.

A radiology caption may mention the modality, anatomical region, imaging plane, abnormality, device, or procedure. These elements are often represented by specialized terminology. Therefore, general captioning models may produce fluent text while still failing to preserve medically important details.

This aspect is important for the present project because medical image caption generation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, medical image caption generation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats medical image caption generation as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.2 UMLS Concepts and Concept-Based Representation

The Unified Medical Language System provides a controlled biomedical vocabulary. In the ImageCLEFmedical Caption context, UMLS concepts are represented by Concept Unique Identifiers. A CUI is a symbolic code, whereas the associated UMLS term provides a human-readable medical name.

Using concepts as intermediate representations can make caption generation more interpretable. A concept set indicates what medical entities are expected to appear in the caption. However, the usefulness of this representation depends on concept quality. Ground-truth concepts provide an upper-bound setting, while predicted concepts reflect the practical deployment scenario.

This aspect is important for the present project because UMLS concept representation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, UMLS concept representation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats UMLS concept representation as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.3 Transformer Encoder-Decoder Models

Transformer encoder-decoder models use self-attention to represent input sequences and cross-attention to generate output sequences. For caption generation, the encoder reads the prompt and the decoder generates the caption token by token. This architecture is well suited for text-to-text generation because it can condition each generated token on the full input representation and previously generated tokens.

This aspect is important for the present project because transformer encoder-decoder models affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, transformer encoder-decoder models must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats transformer encoder-decoder models as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration

succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.4 T5 for Text-to-Text Caption Generation

T5 formulates language tasks as text-to-text transformations. This is appropriate for the present thesis because UMLS concepts and image-derived visual prefix tokens can be converted into an encoder input, while the output remains a medical caption. T5-base provides a practical balance between capacity, availability, and computational feasibility in Colab-based experimentation.

This aspect is important for the present project because T5 text-to-text generation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, T5 text-to-text generation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats T5 text-to-text generation as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.5 Vision-Language Pretraining and BiomedCLIP

Vision-language pretraining learns aligned image and text representations. BiomedCLIP extends this idea to the biomedical domain by using biomedical image-text pairs. In this project, BiomedCLIP is used as a frozen image encoder to obtain visual features that can guide caption generation without training a full vision-language model from scratch.

This aspect is important for the present project because BiomedCLIP visual representations affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, BiomedCLIP visual representations must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric

may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats BiomedCLIP visual representations as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.6 Visual Prefix Learning

Visual prefix learning maps image features into a sequence of learned pseudo-tokens that can be prepended to a language model input. Instead of changing the full language model architecture, a projection adapter converts image embeddings into the same hidden dimension used by T5. This allows visual information to enter the encoder alongside concept text.

This aspect is important for the present project because visual prefix learning affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, visual prefix learning must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats visual prefix learning as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.7 Predicted-Concept Noise and Error Propagation

Predicted concepts may contain false positives and false negatives. False negatives can remove important semantic information from the prompt, while false positives can introduce irrelevant concepts that mislead the generator. Error propagation is therefore a central theoretical issue in concept-guided caption generation.

This aspect is important for the present project because predicted-concept noise affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the

visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, predicted-concept noise must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats predicted-concept noise as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

2.8 Evaluation Metrics for Caption Generation

Caption evaluation requires multiple metrics. ROUGE-1, ROUGE-2, and ROUGE-L measure lexical overlap. BLEU measures precision-oriented n-gram overlap. BERTScore and BLEURT measure semantic similarity. Dictionary-based UMLS-F1 measures internal medical concept consistency by matching UMLS-related terms in generated and reference captions.

$$Precision = TP / (TP + FP)$$

Equation (2.1): Precision. 9

$$Recall = TP / (TP + FN)$$

Equation (2.2): Recall. 9

$$F1 = 2 \times Precision \times Recall / (Precision + Recall)$$

Equation (2.3): F1-score. 9

$$L = - \sum_t \log P(y_t | y_{<t}, x)$$

Equation (2.4): Cross-entropy sequence loss. 9

$$Dict-UMLS-F1 = 2 \times P_{umls} \times R_{umls} / (P_{umls} + R_{umls})$$

Equation (2.5): Dictionary-based UMLS-F1. 9

The official ImageCLEFmedical Caption 2026 evaluation uses relevance and factuality metrics, including BERTScore, ROUGE, BLEURT, UMLS concept F1, and AlignScore. In this thesis, the internal evaluation does not claim to reproduce the official leaderboard exactly; instead, it uses a controlled validation subset for systematic comparison among experiments.

2.9 Chapter Summary

This chapter introduced the theoretical background of concept-guided caption generation, including UMLS concepts, T5, BiomedCLIP, visual prefix learning, concept noise, and evaluation metrics.

Chapter 3: Related Work

This chapter reviews previous work on medical caption generation, concept-guided generation, medical vision-language pretraining, and ImageCLEFmedical Caption systems. It uses the revised literature review prepared for this project as the main basis and expands it into a thesis-level discussion.

3.1 Early Medical Captioning and Tag-Guided Generation

A major lesson from early tag-guided captioning is that semantic labels can reduce the ambiguity of the visual input. When a system knows that an image is related to a specific organ, modality, or finding, the language decoder has a narrower and more clinically meaningful search space. However, early systems often treated tags as auxiliary signals rather than as a central object of analysis. This thesis differs by making concept guidance itself the main experimental variable.

The limitation of early systems is especially visible when medical language must be precise. A general description may sound fluent while still missing the most important diagnostic term. In medical captioning, the absence of a single anatomical or pathological concept can change the meaning of the generated caption. This motivates an approach that measures not only text overlap but also concept-level consistency.

From the perspective of the present project, tag-guided generation provides historical support for the use of intermediate semantics. At the same time, the limitations of those systems justify the transition toward transformer-based generation and standardized UMLS concepts. The use of UMLS allows the project to discuss medical concepts in a controlled and reproducible manner rather than using arbitrary tags.

Early medical captioning approaches often combined convolutional visual encoders with recurrent language decoders. Some methods incorporated tags or semantic labels to improve content selection. These systems established the importance of intermediate semantic cues, but they were limited by weaker language modeling capacity and by the lack of robust factuality control.

This aspect is important for the present project because early tag-guided medical captioning affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, early tag-guided medical captioning must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric

may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats early tag-guided medical captioning as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

This aspect is important for the present project because early tag-guided medical captioning affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

3.2 Transformer-Based Radiology Report Generation

Transformer-based radiology report generation improved the ability of models to capture long-range dependencies and to generate coherent clinical text. Nevertheless, many report-generation models are optimized for longer reports, whereas ImageCLEFmedical Caption focuses on concise captions. This difference matters because short captions must select the most salient information with minimal redundancy.

Another distinction is that many report-generation systems do not use UMLS concepts as explicit control variables. They may implicitly learn clinical semantics from image-report pairs, but the generated output is not necessarily traceable to a specific set of medical concepts. The present thesis takes the opposite direction: it makes the concept set visible and uses it to study the relationship between semantic guidance and generated text.

This controlled formulation makes the experimental comparison more interpretable. If a model performs well with ground-truth concepts but degrades with predicted concepts, the source of the degradation can be linked to upstream concept errors. This type of analysis is difficult in direct image-to-text systems where the intermediate semantic representation is hidden.

Transformer-based report generation models improved long-range language modeling and context handling. Systems such as memory-driven Transformer approaches showed that transformer decoders can generate more coherent radiology text than earlier recurrent methods. However, full report generation differs from short ImageCLEF-style caption prediction, and the use of explicit UMLS concepts is not always central in these models.

This aspect is important for the present project because transformer-based radiology generation affects not only the numerical score but also the interpretability of the generated medical caption. In

concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, transformer-based radiology generation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats transformer-based radiology generation as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

This aspect is important for the present project because transformer-based radiology generation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

3.3 Contrastive Medical Vision-Language Learning

Contrastive pretraining methods such as ConVIRT, MedCLIP, and BiomedCLIP learn aligned representations between images and texts. These models are useful for retrieval, classification transfer, similarity measurement, and feature extraction. In this thesis, BiomedCLIP is used as a frozen visual encoder rather than as a complete caption generator.

This aspect is important for the present project because contrastive medical vision-language learning affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, contrastive medical vision-language learning must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats contrastive medical vision-language learning as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific

configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

This aspect is important for the present project because contrastive medical vision-language learning affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

3.4 Concept-Guided ImageCLEFmedical Caption Systems

ImageCLEFmedical Caption systems provide the closest comparison point because they share the same broad benchmark family and report several common metrics. The 2024 and 2025 systems show an increasing interest in concept-aware generation, prompt engineering, visual-language models, and reranking. The 2026 official leaderboard further shows that high-performing systems combine large foundation models, visual evidence, and factuality-oriented evaluation.

However, official leaderboard scores are not always directly comparable with internal validation experiments. The test set is hidden, the official evaluation uses specific scripts, and the official UMLS metric is based on MedCAT. Therefore, this thesis uses leaderboard values for contextual comparison while clearly separating them from internal validation scores. This distinction protects the scientific validity of the claims.

The most important comparison is not only whether one number is larger than another. The key question is what each system demonstrates. The oracle Exp6 setting demonstrates the potential value of accurate UMLS concepts, while the predicted-concept Exp6 setting demonstrates the practical value of fusing visual features with automatically detected concepts.

Recent ImageCLEFmedical Caption systems increasingly combine visual models, predicted UMLS concepts, prompting, reranking, and post-processing. This trend supports the central direction of the thesis: concept guidance is valuable, but its effect must be analyzed carefully because predicted concepts are noisy and incomplete.

The literature review identified that many systems report strong text similarity scores, but fewer studies isolate the difference between ground-truth concepts and predicted concepts. This thesis directly addresses that gap by evaluating both oracle and practical settings and by performing error analysis on predicted-concept failures.

This aspect is important for the present project because ImageCLEF concept-guided systems affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the

input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, ImageCLEF concept-guided systems must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats ImageCLEF concept-guided systems as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

3.5 Structured Summary of Key Related Studies

Table 3.1: Adapted structured summary of key related studies. 15

Study	Domain	Main contribution	Reported evidence	Relevance to this thesis
Bodenreider (2004)	Biomedical terminology	Introduced UMLS as a unified biomedical terminology system.	Terminology integration	Provides the semantic foundation for CUI-based guidance.
ROCOv2 (2024)	Radiology dataset	Provides radiology images, captions, and metadata for ImageCLEF tasks.	Dataset scale and diversity	Defines the dataset context behind the current experiments.
Jing et al. (2018)	Medical report generation	Used tags, co-attention, and hierarchical generation.	Tag-guided generation	Supports the value of semantic labels for language generation.
R2Gen (2020)	Radiology report generation	Introduced memory-driven Transformer report generation.	Transformer generation	Motivates transformer-based caption generation.
ConVIRT / MedCLIP	Contrastive pretraining	Aligned medical images and text in a shared space.	Image-text alignment	Motivates using biomedical image encoders.
BiomedCLIP	Biomedical VLM	Pretrained on large biomedical image-text pairs.	Biomedical visual features	Used as frozen visual encoder in Exp5 and Exp6.
DS@BioMed 2024	ImageCLEF captioning	Integrated concept detection into caption generation.	BERTScore = 0.6059 val	Closest prior direction for concept-aware captioning.
AUEB NLP Group 2025	ImageCLEF captioning	Used VLMs, UMLS concepts, decoding and reranking.	BERT=0.5884, R-1=0.2176	Strong concept-aware reference system.
CS_Morgan 2025	ImageCLEF captioning	Explored small VLMs and modality-guided prompting.	BERT=0.5180, R-1=0.1598	Relevant for resource-aware practical systems.
AI Stat Lab 2025	ImageCLEF captioning	Used modular captioning and UMLS-related objectives.	BERT=0.5823, R-1=0.2440	High reference threshold in several metrics.
DS@GT ARC 2026	Official 2026 systems	Used foundation-model scaling for caption prediction.	Overall=0.3571 for ranked submission	Context for current-year official leaderboard.

Table 3.1 adapts the large related-work table into a thesis-friendly format. It preserves the important elements: the study, its domain, its contribution, the available evidence or metrics, and its direct relevance to the proposed concept-guided generation pipeline.

3.6 Comparative Analysis and Research Gap

Table 3.2: Research gaps identified from the related literature. 15

Direction	Limitation / Gap	Response in this thesis
Early CNN-RNN systems	Limited language modeling and weak factual control.	Use T5-based generation with explicit concept guidance.
Report-generation models	Often target long reports rather than concise	Adapt generation to short caption prediction

Direction	Limitation / Gap	Response in this thesis
	ImageCLEF captions.	with controlled prompts.
Contrastive VLMs	Strong alignment but not complete caption generators.	Use BiomedCLIP as visual feature source in a fusion architecture.
Concept-guided systems	Concept errors propagate to generated captions.	Compare GT concepts, predicted concepts, and error breakdown.
BLEU/ROUGE-only evaluation	Lexical overlap does not fully capture medical correctness.	Use semantic and dictionary-based UMLS consistency metrics.
Official leaderboard comparison	Official systems use test-set metrics; internal experiments use validation subsets.	Report contextual comparison with clear methodological cautions.

The research gap is therefore clear: previous systems show the usefulness of concepts and strong language models, but the effect of concept quality on downstream caption generation is not isolated in a controlled way. This thesis explicitly compares raw CUI inputs, UMLS term inputs, ground-truth concepts, predicted concepts, image-only generation, and image-concept fusion.

This aspect is important for the present project because the research gap and proposed innovation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, the research gap and proposed innovation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats the research gap and proposed innovation as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

This aspect is important for the present project because the research gap and proposed innovation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

3.7 Chapter Summary

This chapter reviewed the literature and identified the gap addressed by the thesis. The key insight is that concept guidance is promising, but its practical value depends on how robust the caption generator is to predicted-concept errors.

Chapter 4: Research Methodology

This chapter describes the proposed methodology in detail. It explains the research design, dataset organization, concept mapping, prompt construction, experimental progression, image-concept fusion model, training configuration, evaluation protocol, and error-analysis pipeline.

4.1 Overall Research Design

The research follows an incremental experimental methodology. Each experiment answers a specific question and motivates the next stage. The pipeline begins with concept-only text generation, then introduces UMLS term enrichment, predicted-concept robustness, mixed training, image-only visual input, and finally image-concept fusion.

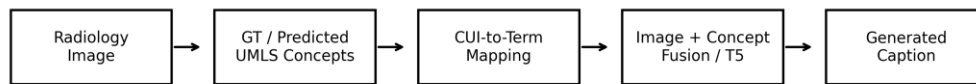


Figure 4.1: Concept-guided caption generation pipeline. 18

Figure 4.1 summarizes the full research pipeline from radiology image and UMLS concept inputs to final generated caption evaluation.

This aspect is important for the present project because overall research design affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, overall research design must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats overall research design as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

4.2 Dataset and Data Organization

The data organization stage was designed to keep all experimental settings comparable. Each validation sample is linked to the same reference caption, while only the input representation changes across experiments. This prevents the comparison from being confounded by different validation splits or different target captions.

The distinction between ground-truth concepts and predicted concepts is central to the methodology. Ground-truth concepts are used as an oracle semantic source because they represent the information that the generator would ideally receive. Predicted concepts, on the other hand, come from the previous concept detection model and therefore include the realistic errors that would appear in a deployable pipeline.

For the image-based experiments, images are not passed directly to T5. Instead, a frozen biomedical image encoder extracts visual features, and a trainable adapter transforms these features into prefix tokens. This design is computationally practical and avoids the need to fully fine-tune a large multimodal foundation model from scratch.

The experiments use the ImageCLEFmedical Caption 2026 development data prepared for caption generation. Each sample is linked to an image identifier, a reference caption, ground-truth UMLS concepts, and predicted UMLS concepts produced by the previous concept detection model. The validation evaluation is performed on an internal subset of 3,000 generated captions per setting.

Table 4.1: Dataset and input resources used in the project. 19

Resource	Description	Role in the experiments
Images	Radiology images from the development split	Used in Exp5 and Exp6 through BiomedCLIP features.
Reference captions	Gold textual descriptions	Used as targets for training and evaluation.
GT CUIs	Ground-truth UMLS concept identifiers	Used for oracle upper-bound experiments.
Pred micro concepts	Micro-optimized predicted concept set	Practical clean concept setting.
Pred coverage concepts	Coverage-oriented predicted concept set	Practical higher-recall concept setting.
CUI-to-term map	Mapping from concept identifiers to readable medical terms	Transforms symbolic CUIs into natural-language prompts.

This aspect is important for the present project because dataset organization affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, dataset organization must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still

introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats dataset organization as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

4.3 Concept Mapping and Prompt Construction

Prompt construction is treated as part of the model design. The same model architecture can behave very differently depending on whether the input is an opaque list of CUIs, a readable list of UMLS terms, or a multimodal sequence containing both visual prefix tokens and concept text. Therefore, each prompt format is evaluated as an explicit experimental condition.

The UMLS term-enriched prompt makes the semantic content accessible to the language model. This is important because T5 was pretrained on natural language rather than on isolated CUI identifiers. By converting CUI codes into terms, the prompt becomes closer to the language distribution that T5 can interpret.

In predicted-concept settings, the prompt may include missing or noisy terms. Rather than hiding this problem, the thesis makes it measurable. The downstream caption metrics are compared with prompt precision, prompt recall, and prompt F1 to explain how upstream concept quality affects generation.

Raw UMLS CUIs are not naturally meaningful to a general language model. Therefore, a key preprocessing step is mapping each CUI to a readable UMLS term. The prompt is then constructed as a structured list of medical terms. This step turns symbolic concept detection outputs into natural-language guidance that T5 can understand.

The mapping strategy is also important for predicted concepts. The same mapping is applied to ground-truth, predicted-micro, and predicted-coverage concept sets, ensuring that the comparison isolates concept quality rather than differences in prompt formatting.

This aspect is important for the present project because concept mapping and prompt construction affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, concept mapping and prompt construction must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one

metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats concept mapping and prompt construction as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

4.4 Experimental Progression

Table 4.2: Summary of the experimental progression from Exp1 to Exp6. 21

Experiment	Input / Training Design	Purpose
Exp1	GT raw CUIs only	Test whether raw symbolic CUIs are sufficient for T5 caption generation.
Exp2	GT CUI + UMLS terms	Measure the value of converting CUIs into readable medical terms.
Exp3	Predicted-only fine-tuning	Improve robustness to noisy predicted concepts.
Exp4	Mixed GT + predicted training	Balance oracle semantic quality and predicted-concept robustness.
Exp5	Image-only visual prefix	Measure the contribution of image features without concept text.
Exp6	Image + concept fusion	Combine visual evidence with UMLS concept guidance.

4.4.1 Exp1: Raw CUI-Only Caption Generation

Exp1 uses only raw CUI identifiers as input. This experiment serves as a symbolic baseline and tests whether T5 can learn useful captioning behavior from opaque concept codes.

This aspect is important for the present project because 4.4.1 Exp1: Raw CUI-Only Caption Generation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

4.4.2 Exp2: UMLS Term-Enriched Caption Generation

Exp2 adds readable UMLS names to the input. This tests whether semantic enrichment of the prompt improves generation quality compared with raw CUIs.

This aspect is important for the present project because 4.4.2 Exp2: UMLS Term-Enriched Caption Generation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

4.4.3 Exp3: Predicted-Concept Fine-Tuning

Exp3 fine-tunes the model using predicted concepts. Its purpose is to reduce the mismatch between oracle training and practical predicted-concept inference.

This aspect is important for the present project because 4.4.3 Exp3: Predicted-Concept Fine-Tuning affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

4.4.4 Exp4: Mixed Ground-Truth and Predicted Training

Exp4 mixes ground-truth and predicted concepts during training. This design aims to keep the model grounded in correct semantics while improving robustness to realistic noise.

This aspect is important for the present project because 4.4.4 Exp4: Mixed Ground-Truth and Predicted Training affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

4.4.5 Exp5: Image-Only Visual Prefix Baseline

Exp5 introduces image features through BiomedCLIP and a trainable visual prefix adapter, without concept text. It measures the value of visual evidence alone.

This aspect is important for the present project because 4.4.5 Exp5: Image-Only Visual Prefix Baseline affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

4.4.6 Exp6: Image-Concept Fusion

Exp6 combines BiomedCLIP visual prefix tokens with UMLS concept text. This is the final proposed fusion model and the strongest architecture in the thesis.

This aspect is important for the present project because 4.4.6 Exp6: Image-Concept Fusion affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the

input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

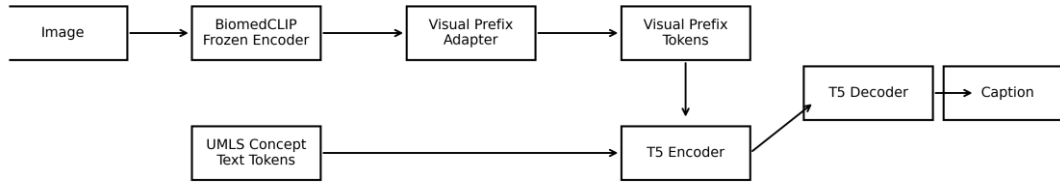


Figure 4.2: Exp6 image-concept fusion architecture. 23

Figure 4.2 shows how frozen biomedical visual features and UMLS concept text are combined before T5 decodes the final caption.

4.5 Training Configuration and Implementation Environment

The experiments were implemented in Google Colab using GPU acceleration. T5-base was used for text generation experiments, while BiomedCLIP provided frozen image features for Exp5 and Exp6. Mixed precision was used when appropriate, and generation was evaluated on a fixed validation subset to preserve comparability among models.

This aspect is important for the present project because training configuration and implementation environment affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, training configuration and implementation environment must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats training configuration and implementation environment as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

This aspect is important for the present project because training configuration and implementation environment affects not only the numerical score but also the interpretability of the generated

medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

4.6 Evaluation Protocol

The evaluation protocol deliberately combines lexical, semantic, and concept-level metrics. ROUGE and BLEU are useful because they reveal overlap with the reference wording. BERTScore and BLEURT are useful because they capture semantic similarity beyond exact word matches. Dictionary-based UMLS-F1 is useful because it focuses on medical concept preservation.

No single metric is sufficient for medical caption generation. A model can obtain a good semantic score while omitting a critical finding, or it can match several words while describing the wrong anatomy. For this reason, the thesis interprets the metrics together and pays particular attention to cases where the metrics disagree.

The comparison with the official 2026 leaderboard is framed carefully. The official task uses a hidden test set and computes an overall score from several relevance and factuality metrics. The present thesis reports internal validation results and therefore uses the leaderboard only as a contextual reference, not as an official ranking claim.

The evaluation protocol reports ROUGE-1, ROUGE-2, ROUGE-L, BLEU, BERTScore-F1, BLEURT, and dictionary-based UMLS-F1. The official 2026 leaderboard includes additional metrics such as image-caption similarity, MedCAT-based UMLS concept F1, and AlignScore. Therefore, leaderboard comparison is used for context rather than as a direct official submission claim.

This aspect is important for the present project because evaluation protocol affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, evaluation protocol must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats evaluation protocol as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

4.7 Error Analysis Protocol

The error analysis computes per-sample lexical and medical concept scores, missed reference UMLS concepts, hallucinated generated UMLS concepts, prompt precision, prompt recall, and prompt F1. This allows the thesis to explain not only which model performs best, but also why errors occur.

This aspect is important for the present project because error analysis protocol affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, error analysis protocol must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats error analysis protocol as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

4.8 Methodological Limitations

The most important methodological limitation is that the experiments are evaluated on an internal validation subset rather than the official hidden test leaderboard for Senior Project II. In addition, dictionary-based UMLS-F1 is an internal consistency proxy and is not guaranteed to be identical to the official MedCAT-based UMLS metric.

This aspect is important for the present project because methodological limitations affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, methodological limitations must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats methodological limitations as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

4.9 Chapter Summary

This chapter presented the complete research methodology. The staged design allows the thesis to isolate the contributions of UMLS term enrichment, predicted-concept robustness, image-only evidence, and final image-concept fusion.

Chapter 5: Experiments and Results

This chapter reports the quantitative results of the six experimental stages. It interprets the final metrics, compares oracle and practical settings, and contextualizes the results against previous references and the official ImageCLEFmedical Caption 2026 leaderboard.

5.1 Experimental Settings

All experiments are evaluated on the same internal validation subset of 3,000 generated captions per setting. This ensures fair comparison among the proposed configurations. The metrics are computed consistently across Exp1-Exp6, with corrected non-rescaled BERTScore values used for final reporting.

This aspect is important for the present project because experimental settings affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, experimental settings must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

5.2 Final Exp1-Exp6 Results

Table 5.1: Final Exp1-Exp6 results on the internal validation subset. 27

Exp	Setting	Input	R-1	R-2	R-L	BLEU	BERT-F1	BLEURT	Dict-UMLS F1
Exp1	GT raw CUIs only	Text-only	0.1787	0.0512	0.1516	2.4034	0.5333	0.2873	0.0989
Exp2	GT concepts + UMLS terms	Text-only / GT Concepts	0.2763	0.1123	0.2452	5.1341	0.5887	0.3133	0.4694
Exp2	Pred micro concepts + UMLS terms	Text-only / Predicted Concepts	0.1454	0.0430	0.1306	1.2092	0.4738	0.2281	0.2491
Exp2	Pred coverage concepts + UMLS terms	Text-only / Predicted Concepts	0.1446	0.0448	0.1291	1.2210	0.4828	0.2324	0.2599
Exp3	GT concepts + UMLS terms	Text-only / GT Concepts	0.2880	0.1068	0.2476	5.0203	0.6066	0.3286	0.3794
Exp3	Pred micro concepts + UMLS terms	Text-only / Predicted Concepts	0.2083	0.0633	0.1807	2.2762	0.5925	0.2924	0.1765
Exp3	Pred coverage concepts + UMLS terms	Text-only / Predicted Concepts	0.2114	0.0657	0.1831	2.2997	0.5915	0.2937	0.1882
Exp4	GT concepts + UMLS terms	Text-only / GT Concepts	0.3328	0.1414	0.2949	5.6574	0.6405	0.3469	0.4972
Exp4	Pred micro concepts + UMLS terms	Text-only / Predicted Concepts	0.2179	0.0708	0.1889	2.5947	0.5716	0.2862	0.2130
Exp4	Pred coverage concepts + UMLS terms	Text-only / Predicted Concepts	0.2185	0.0718	0.1897	2.5289	0.5727	0.2876	0.2235
Exp5	Image-only	Image-only	0.2401	0.0806	0.2070	2.8757	0.6043	0.3385	0.3521
Exp6	Image + GT	Image + GT	0.3656	0.1582	0.3205	6.6925	0.6656	0.4197	0.5850

Exp	Setting	Input	R-1	R-2	R-L	BLEU	BERT-F1	BLEURT	Dict-UMLS F1
	concepts	Concepts							
Exp6	Image + Pred micro concepts	Image + Predicted Concepts	0.2504	0.0867	0.2159	3.5394	0.6019	0.3463	0.4088
Exp6	Image + Pred coverage concepts	Image + Predicted Concepts	0.2513	0.0870	0.2168	3.4938	0.6028	0.3460	0.4129

Table 5.1 shows that the best overall result is obtained by Exp6 with image features and ground-truth UMLS concepts. This setting is highlighted because it represents the strongest oracle upper-bound and numerically exceeds all other internal configurations across the reported metrics.

Table 5.2: Best metric values achieved by the proposed configurations. 28

Metric	Best setting	Best value
ROUGE-1	Exp6 Image + GT concepts	0.3656
ROUGE-2	Exp6 Image + GT concepts	0.1582
ROUGE-L	Exp6 Image + GT concepts	0.3205
BLEU	Exp6 Image + GT concepts	6.6925
BERTScore-F1	Exp6 Image + GT concepts	0.6656
BLEURT	Exp6 Image + GT concepts	0.4197
Dict-UMLS F1	Exp6 Image + GT concepts	0.5850

5.3 Metric-by-Metric Interpretation

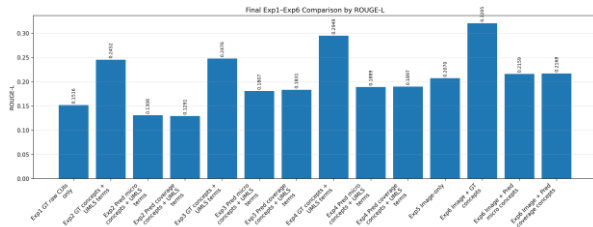


Figure 5.1: Final Exp1-Exp6 ROUGE-L comparison.

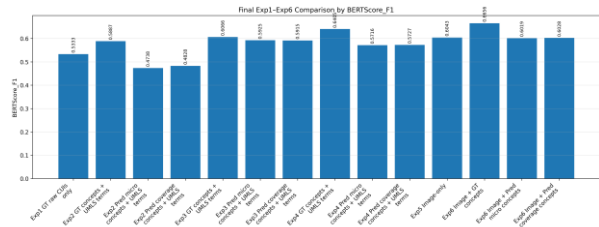


Figure 5.2: Final Exp1-Exp6 BERTScore-F1 comparison.

Figure 5.1 shows the progression from raw CUI input to the final fusion model; ROUGE-L reaches its maximum in the Exp6 oracle setting. Figure 5.2 indicates that semantic similarity also improves most clearly when visual information and correct concepts are fused.

The lexical metrics show that raw CUIs are insufficient for strong caption generation, while UMLS term enrichment improves performance substantially. The semantic metrics show that image-based features are useful, but the highest scores require both image evidence and correct semantic concepts.

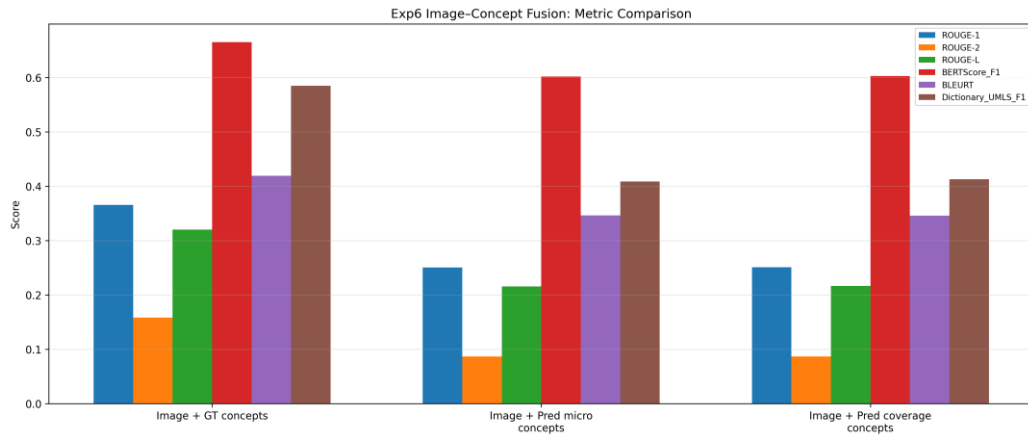


Figure 5.3: Exp6 fusion metric comparison. 29

Figure 5.3 compares the three Exp6 settings and shows the gap between oracle ground-truth concepts and practical predicted concepts.

This aspect is important for the present project because metric-by-metric interpretation affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, metric-by-metric interpretation must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

5.4 Oracle Upper-bound versus Practical Settings

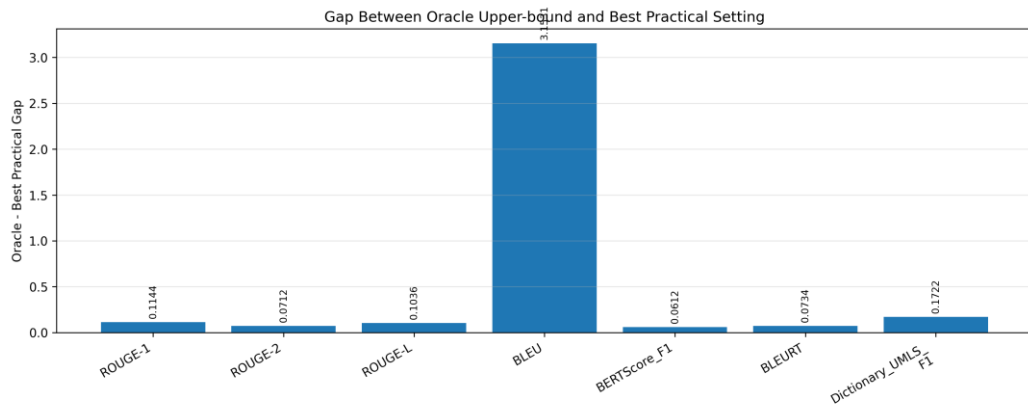


Figure 5.4: Oracle upper-bound versus best practical setting gap. 29

Figure 5.4 highlights the remaining performance gap caused mainly by incomplete or noisy predicted concepts.

The oracle setting should not be interpreted as a practical deployed system because it uses ground-truth concepts. However, it is scientifically useful because it shows the potential value of accurate concept guidance. The practical settings use predicted concepts and therefore represent the realistic pipeline.

This aspect is important for the present project because oracle versus practical analysis affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, oracle versus practical analysis must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

5.5 Comparison with Previous Reference Systems

Table 5.3: Official ImageCLEFmedical Caption 2026 leaderboard subset. 30

Rank	Team	Overall	BERT	ROUGE	BLEURT	MedCAT	Align
1	AIstatLab	0.3755	0.6250	0.3234	0.3555	0.2160	0.1287
2	UFPE-Essex-UNED	0.3603	0.6017	0.2725	0.3463	0.1822	0.1537
3	dsgt_caption	0.3571	0.6042	0.2809	0.3229	0.1943	0.1610
4	AUEB/DMST	0.3516	0.6130	0.2728	0.3183	0.1812	0.1674
5	csmorgan	0.3458	0.6075	0.2725	0.3299	0.1751	0.1327

Table 5.4: Focused comparison between our best results, references, and 2026 systems.

30

System	Comparison type	Reported comparable values	Interpretation
Our Exp6 oracle	Image + GT concepts	R-1=0.3656, BERT=0.6656, BLEURT=0.4197, Dict-UMLS=0.5850	Numerically higher than the shown 2026 leaderboard values on shared metrics; reported only as oracle upper-bound.
Our Exp6 practical	Image + Pred coverage	R-1=0.2513, BERT=0.6028, BLEURT=0.3460, Dict-UMLS=0.4129	Competitive with high-ranking 2026 systems, especially in BERT/BLEURT, but lower than Rank 1 in ROUGE and BERT.
DS@BioMed 2024	Reference	BERT=0.6059 validation	Strong 2024 concept-aware baseline.
AUEB NLP Group 2025	Reference	R-1=0.2176, BERT=0.5884, UMLS=0.1429	Our practical fusion is higher on R-1, BERT, and internal Dict-UMLS.
AI Stat Lab 2025	Reference	R-1=0.2440, BERT=0.5823, BLEURT=0.3173, UMLS=0.1524	Our practical fusion is higher on R-1, BERT, BLEURT, and internal Dict-UMLS.
AIstatLab 2026 Rank 1	Official leaderboard	Overall=0.3755, BERT=0.6250, ROUGE=0.3234, BLEURT=0.3555	Top official system; our practical result does not exceed it overall.
UFPE-Essex-UNED 2026 Rank 2	Official leaderboard	BERT=0.6017, ROUGE=0.2725, BLEURT=0.3463	Our practical model slightly exceeds BERT and approximately matches

System	Comparison type	Reported comparable values	Interpretation
			BLEURT, but is lower in ROUGE.
AUEB/DMST 2026 Rank 4	Official leaderboard	BERT=0.6130, ROUGE=0.2728, BLEURT=0.3183	Our practical model is higher in BLEURT but lower in BERT and ROUGE.

Table 5.4 is the central comparison table of the thesis. It deliberately highlights the strongest Exp6 oracle result while also separating it from the practical predicted-concept fusion setting. This structure shows the numerical strength of accurate concept guidance without making an unfair official leaderboard claim.

5.6 Comparison with the Official 2026 Leaderboard

The official 2026 Caption Prediction leaderboard uses an overall score computed from multiple relevance and factuality metrics. Therefore, our internal validation results should be interpreted as contextual evidence rather than official leaderboard ranks. Nevertheless, the comparison is useful because it shows that the proposed method is competitive with high-ranking 2026 systems on several shared metrics.

Compared with the second-ranked 2026 system, the practical Exp6 predicted-coverage setting is slightly higher in BERTScore-F1, while the predicted-micro setting approximately matches BLEURT. Compared with the fourth-ranked system, the practical Exp6 setting is higher in BLEURT but lower in BERT and ROUGE. The oracle Exp6 setting is numerically higher than the shown leaderboard values on shared metrics, but it uses ground-truth concepts and must be described as an upper-bound.

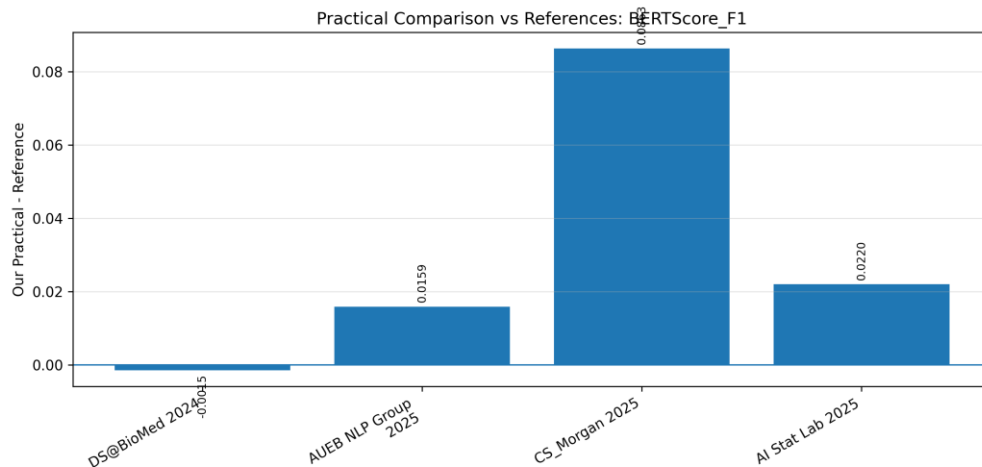


Figure 5.5: Practical comparison against selected references. 31

Figure 5.5 shows that practical predicted-concept models are competitive against several reference systems, while the oracle setting remains the strongest numerical upper-bound.

This aspect is important for the present project because official 2026 leaderboard comparison affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, official 2026 leaderboard comparison must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

5.7 Discussion of Main Findings

The main experimental trend is consistent across metrics: semantic guidance improves caption generation, but the quality of that guidance matters. Raw CUIs provide weak input because they are symbolic identifiers. Readable UMLS terms are substantially more useful because they expose medical meaning to the language model. Image features provide additional grounding, especially when concept prompts are incomplete.

The strongest numerical result is the Exp6 oracle setting. This is an important finding because it demonstrates that the proposed architecture can use accurate concepts effectively when they are available. The result should be highlighted in the thesis because it shows the upper-bound potential of concept-guided fusion. At the same time, the thesis must state that this setting is not a practical deployment scenario.

The practical predicted-concept fusion results show a more realistic picture. They do not exceed the top official 2026 system overall, but they are competitive with high-ranking systems on selected shared metrics. More importantly, they improve internal medical concept consistency compared with the text-only predicted-concept baselines, which supports the value of the fusion design.

The results support three main findings. First, readable UMLS terms are much more useful than raw CUI identifiers for T5-based generation. Second, predicted-concept robustness is necessary because a model trained only on ground-truth concepts suffers when evaluated with predicted concepts. Third, image-concept fusion provides the best overall behavior because it combines visual evidence with structured medical semantic guidance.

The practical fusion model does not officially outperform the top 2026 leaderboard system, but it is competitive and it demonstrates a clear research contribution. The strongest scientific claim is that accurate concept guidance has a large upper-bound effect and that predicted-concept fusion can preserve part of that benefit in realistic settings.

This aspect is important for the present project because main findings affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, main findings must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats main findings as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

5.8 Chapter Summary

This chapter presented the final results and comparisons. Exp6 Image + GT concepts achieved the best overall internal result and illustrates the upper-bound potential of accurate UMLS concept guidance. Exp6 with predicted concepts represents the practical system and remains competitive with several reference systems.

Chapter 6: Error Analysis and Discussion

This chapter analyzes why different configurations succeed or fail. It focuses on missed concepts, hallucinated concepts, prompt quality, and the difference between predicted micro and predicted coverage concept sources.

6.1 Purpose of the Error Analysis

The purpose of error analysis is to move beyond aggregate metrics. A high BERTScore may hide medical omissions, while a reasonable ROUGE score may still correspond to an anatomically incorrect caption. Therefore, per-sample medical concept analysis is essential for understanding the behavior of the proposed system.

This aspect is important for the present project because purpose of the error analysis affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, purpose of the error analysis must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats purpose of the error analysis as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

6.2 Error Metrics and Analysis Tables

Table 6.1: Error breakdown summary for the main models. 34

Model / Setting	Mean sample ROUGE-L	Mean Dict-UMLS F1	Interpretation
Exp3 GT	0.2391	0.3104	Oracle concept setting
Exp3 pred coverage	0.1789	0.1370	Higher prompt recall but more noisy concepts
Exp3 pred micro	0.1775	0.1259	Cleaner prompts but slightly lower downstream concept F1
Exp4 GT	0.2832	0.4245	Strong text-only oracle baseline
Exp4 pred coverage	0.1840	0.1349	Improves lexical quality but concept consistency remains limited
Exp5 image-only	0.1986	0.1820	Visual evidence helps even without concepts
Exp6 Image + GT	0.3078	0.4442	Best error-analysis behavior
Exp6 Image + Pred coverage	0.2077	0.1826	Best practical error-analysis setting
Exp6 Image + Pred micro	0.2071	0.1800	Very close to coverage, slightly cleaner concept source

Table 6.1 confirms that the oracle fusion model provides the strongest error-analysis behavior. Among practical settings, the image-concept fusion models outperform concept-only settings, showing that visual evidence helps compensate for predicted-concept errors.

6.3 Missed and Hallucinated UMLS Concepts

The missed-concept analysis shows that the generator is strongly influenced by the semantic completeness of the prompt. When the prompt lacks a key anatomical term, the model may choose a more common or more visually plausible description. This is especially problematic in medical captioning because a small anatomical shift can make the caption clinically incorrect.

Hallucinated generated concepts represent the opposite failure mode. In these cases, the model adds medical entities that are not supported by the reference caption. Some hallucinations may be linguistically plausible, but they remain undesirable because they reduce factual reliability. This is why medical captioning must be evaluated with concept-aware metrics, not only with fluency or semantic similarity.

The fusion model partially reduces these errors because visual features provide a second source of evidence. If the concept prompt is incomplete, the image embedding may still contain modality or anatomical cues. However, visual features do not fully solve the problem, which confirms that improving upstream concept detection remains an important direction for future work.

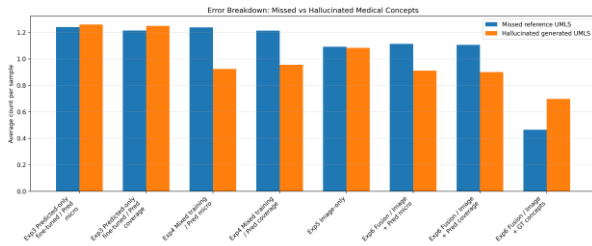


Figure 6.1: Missed versus hallucinated UMLS concepts.

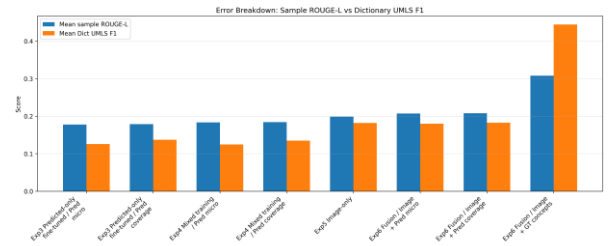


Figure 6.2: Sample ROUGE-L versus dictionary-based UMLS-F1.

Figure 6.1 shows that practical systems still miss important medical concepts, but fusion reduces the downstream effect compared with text-only predicted settings. Figure 6.2 shows that lexical quality and medical concept consistency do not always improve equally, which justifies reporting both metrics.

The most damaging errors occur when the input concept set lacks important anatomical or modality information. In such cases, the generator may infer a plausible but incorrect body region or imaging modality. Hallucinated concepts also matter, but the analysis suggests that missing concepts are often more harmful than extra noisy concepts because missing information cannot easily be recovered by a text-only generator.

This aspect is important for the present project because missed and hallucinated medical concepts affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, missed and hallucinated medical concepts must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats missed and hallucinated medical concepts as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

6.4 Micro versus Coverage Concept Sources

Table 6.2: Predicted micro versus predicted coverage error summary. 36

Setting	Prompt precision	Prompt recall	Missing concepts	Noisy concepts	Mean R-L	Mean Dict-UMLS F1
Exp3 coverage	0.8638	0.5069	1.9133	0.3417	0.1789	0.1370
Exp3 micro	0.9089	0.4975	1.9473	0.1960	0.1775	0.1259
Exp4 coverage	0.8638	0.5069	1.9133	0.3417	0.1840	0.1349
Exp4 micro	0.9089	0.4975	1.9473	0.1960	0.1833	0.1244
Exp6 coverage	0.8638	0.5069	1.9133	0.3417	0.2077	0.1826
Exp6 micro	0.9089	0.4975	1.9473	0.1960	0.2071	0.1800

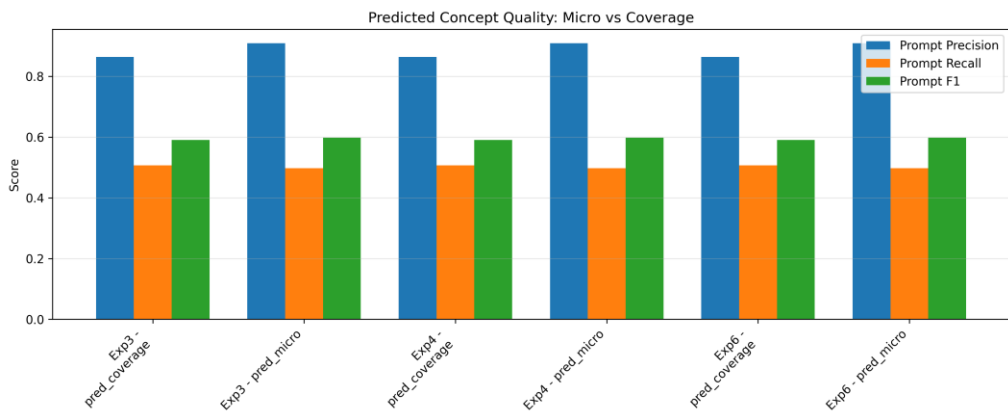


Figure 6.3: Predicted concept quality for micro and coverage settings. 36

Figure 6.3 illustrates the precision-recall trade-off between micro and coverage concept sources.

The micro setting is cleaner and has higher prompt precision, while the coverage setting includes more relevant concepts at the cost of additional noise. In downstream caption generation, coverage often performs slightly better in ROUGE-L and dictionary-based UMLS-F1, suggesting that concept recall is particularly important when generating medically grounded captions.

This aspect is important for the present project because micro versus coverage concept sources affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, micro versus coverage concept sources must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats micro versus coverage concept sources as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

6.5 Qualitative Best and Worst Examples

Qualitative examples support the quantitative findings. The best examples usually involve clear images and concept prompts that contain the correct modality and anatomy, such as simple captions describing brain MRI, chest X-ray, or ultrasound views. The worst examples often involve severe concept omissions that cause the generator to describe the wrong anatomical region or modality.

A representative failure pattern occurs when the reference describes pelvic or abdominal findings but the predicted concept prompt is incomplete. The generated caption may then drift toward thoracic or MRI-related descriptions. This behavior demonstrates that concept-guided generation remains sensitive to upstream concept quality.

This aspect is important for the present project because qualitative examples affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, qualitative examples must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still

introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats qualitative examples as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

6.6 Discussion of Clinical and Research Implications

From a research perspective, the error analysis supports the idea that caption generation should be studied as a pipeline. A downstream language model cannot be evaluated independently from the quality of the upstream concept detector. The oracle-practical gap provides a measurable estimate of how much improvement might be possible if concept prediction becomes more accurate.

From a clinical perspective, the results emphasize caution. The system is not a diagnostic tool, and generated captions should not be used without expert review. Nevertheless, the architecture is valuable as a research prototype because it produces interpretable intermediate signals and allows errors to be traced back to concept-level causes.

The most promising practical direction is to improve concept recall without introducing excessive noise. The coverage setting suggests that additional relevant concepts can help caption generation, but only if the generator is robust enough to ignore irrelevant terms. Future models could use confidence scores, concept grouping, or attention masking to control this trade-off more effectively.

The results have two important implications. First, medical caption generation should be evaluated using concept-aware metrics in addition to general text similarity metrics. Second, improving concept detection remains important even when image features are added, because the concept prompt still controls important semantic aspects of the generated caption.

The image-concept fusion architecture provides a useful compromise. Image features offer visual grounding, while concept text provides explicit medical semantics. This combination is more interpretable than pure image-to-text generation and more robust than concept-only generation under predicted-concept noise.

This aspect is important for the present project because clinical and research implications affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, clinical and research implications must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

The thesis therefore treats clinical and research implications as an analytical dimension. The goal is not only to obtain a high automatic score, but also to explain why a specific configuration succeeds or fails under ground-truth concepts, predicted concepts, and image-concept fusion settings.

This aspect is important for the present project because clinical and research implications affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

6.7 Chapter Summary

This chapter showed that caption quality is strongly affected by concept quality. Exp6 reduces the negative impact of predicted-concept noise, but the remaining oracle-practical gap confirms that upstream concept detection remains a key bottleneck.

Chapter 7: Conclusion and Future Work

7.1 Final Conclusion

This thesis presented a concept-guided approach to medical image caption generation using UMLS concept detection and image-concept fusion. The experimental progression demonstrated that raw CUIs are insufficient, UMLS term enrichment is highly beneficial, predicted-concept fine-tuning improves robustness, and image-concept fusion provides the strongest overall behavior.

The best internal result was achieved by Exp6 using image features and ground-truth UMLS concepts. This result should be interpreted as an oracle upper-bound because it uses concept annotations not available at deployment. The practical predicted-concept settings remained competitive and showed that fusing image features with predicted concepts can improve medical concept consistency compared with text-only predicted-concept baselines.

This aspect is important for the present project because final conclusion affects not only the numerical score but also the interpretability of the generated medical caption. In concept-guided captioning, the model output is influenced by the semantic completeness of the input concepts, the visual evidence encoded by the image representation, and the language model ability to form a concise medically meaningful sentence.

From an engineering perspective, final conclusion must be evaluated as part of a complete pipeline rather than as an isolated modeling choice. A design that improves one metric may still introduce hallucination, reduce concept coverage, or produce captions that are semantically plausible but medically incomplete.

7.2 Main Contributions

- A full experimental chain from concept-only baselines to final image-concept fusion.
- A controlled analysis of ground-truth versus predicted UMLS concept guidance.
- A practical fusion architecture combining BiomedCLIP visual features with T5 concept-guided generation.
- A metric suite covering lexical overlap, semantic similarity, and dictionary-based medical concept consistency.
- An error analysis that links upstream concept quality to downstream caption failures.

7.3 Limitations

- The final captioning experiments were evaluated on an internal validation subset rather than the official hidden test leaderboard.
- Dictionary-based UMLS-F1 is an internal proxy and may differ from official MedCAT-based UMLS scoring.
- The visual encoder was frozen; full multimodal fine-tuning may provide additional gains but requires more compute.
- The generated captions remain short and may not capture all clinical nuance present in longer radiology reports.

7.4 Future Work

- Evaluate the final fusion model on the official hidden test set when possible.
- Integrate MedCAT or official evaluation scripts for closer alignment with challenge factuality metrics.
- Explore larger biomedical language models and instruction-tuned medical VLMs.
- Improve predicted concept calibration to reduce the oracle-practical gap.
- Add uncertainty estimation so generated captions can reflect confidence in predicted concepts.
- Study retrieval-augmented captioning using visually similar images and captions.

The most direct future direction is to reduce the gap between the oracle and practical settings. This requires improving concept detection recall without adding excessive noise, and improving the fusion model so that it can use image evidence to correct incomplete concept prompts.

References

- [1] Bodenreider, O. (2004). The Unified Medical Language System (UMLS): integrating biomedical terminology. *Nucleic Acids Research*, 32(Database issue), D267-D270.
- [2] Pelka, O., Koitka, S., Rückert, J., Nensa, F., & Friedrich, C. M. (2024). ROCov2: Radiology Objects in Context Version 2, an updated multimodal image dataset. *Scientific Data*.
- [3] ImageCLEFmedical Caption 2026. Official task page and leaderboard. ImageCLEF / LifeCLEF, 2026.
- [4] Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2018). ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *CVPR*.
- [5] Irvin, J., et al. (2019). CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. *AAAI*.
- [6] Bustos, A., Pertusa, A., Salinas, J. M., & de la Iglesia-Vayá, M. (2020). PadChest: A large chest X-ray image dataset with multi-label annotated reports. *Medical Image Analysis*.
- [7] Chen, Z., Song, Y., Chang, T.-H., & Wan, X. (2020). Generating radiology reports via memory-driven Transformer. *EMNLP*.
- [8] Zhang, Y., Jiang, H., Miura, Y., Manning, C. D., & Langlotz, C. P. (2020). Contrastive learning of medical visual representations from paired images and text. *MLHC*.
- [9] Wang, Z., Wu, Z., Agarwal, D., & Sun, J. (2022). MedCLIP: Contrastive learning from unpaired medical images and text. *EMNLP*.
- [10] Zhang, S., et al. (2023). BiomedCLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint*.
- [11] Raffel, C., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text Transformer. *Journal of Machine Learning Research*.
- [12] Lin, C.-Y. (2004). ROUGE: A package for automatic evaluation of summaries. *ACL Workshop*.
- [13] Papineni, K., Roukos, S., Ward, T., & Zhu, W.-J. (2002). BLEU: a method for automatic evaluation of machine translation. *ACL*.
- [14] Zhang, T., Kishore, V., Wu, F., Weinberger, K., & Artzi, Y. (2020). BERTScore: Evaluating text generation with BERT. *ICLR*.

- [15] Sellam, T., Das, D., & Parikh, A. (2020). BLEURT: Learning robust metrics for text generation. ACL.
- [16] Zha, Y., et al. (2023). AlignScore: Evaluating factual consistency with a unified alignment function. ACL.
- [17] Nguyen, N. N.-Y., et al. (2024). DS@BioMed at ImageCLEFmedical Caption 2024: Enhanced attention mechanisms in medical caption generation through concept detection integration. Working Notes of CLEF.
- [18] AUEB NLP Group. (2025). ImageCLEFmedical Caption 2025 working notes on concept-aware medical caption generation. Working Notes of CLEF.
- [19] CS_Morgan. (2025). ImageCLEFmedical Caption 2025 working notes on small VLM adaptation and medical caption prediction. Working Notes of CLEF.
- [20] AI Stat Lab. (2025). ImageCLEFmedical Caption 2025 working notes on modular concept-aware caption generation. Working Notes of CLEF.
- [21] Wang, B., Zhang, Y., & Mehta, R. (2026). DS@GT ARC at ImageCLEFmedical 2026: Architectural diversity for concept detection and foundation-model scaling for caption prediction in medical image analysis. arXiv preprint.

Appendix A: Supplementary Experiment Tables

The appendices are kept concise and are intended to support reproducibility and defense discussion. They should remain in the thesis if the supervisor accepts supplementary material; otherwise, they can be removed without changing the main scientific argument.

Table A.1: Supplementary detailed experimental metrics. 44

Exp	Setting	Input	R-1	R-2	R-L	BLEU	BERT-F1	BLEURT	Dict-UMLS F1
Exp1	GT raw CUIs only	Text-only	0.1787	0.0512	0.1516	2.4034	0.5333	0.2873	0.0989
Exp2	GT concepts + UMLS terms	Text-only / GT Concepts	0.2763	0.1123	0.2452	5.1341	0.5887	0.3133	0.4694
Exp2	Pred micro concepts + UMLS terms	Text-only / Predicted Concepts	0.1454	0.0430	0.1306	1.2092	0.4738	0.2281	0.2491
Exp2	Pred coverage concepts + UMLS terms	Text-only / Predicted Concepts	0.1446	0.0448	0.1291	1.2210	0.4828	0.2324	0.2599
Exp3	GT concepts + UMLS terms	Text-only / GT Concepts	0.2880	0.1068	0.2476	5.0203	0.6066	0.3286	0.3794
Exp3	Pred micro concepts + UMLS terms	Text-only / Predicted Concepts	0.2083	0.0633	0.1807	2.2762	0.5925	0.2924	0.1765
Exp3	Pred coverage concepts + UMLS terms	Text-only / Predicted Concepts	0.2114	0.0657	0.1831	2.2997	0.5915	0.2937	0.1882
Exp4	GT concepts + UMLS terms	Text-only / GT Concepts	0.3328	0.1414	0.2949	5.6574	0.6405	0.3469	0.4972
Exp4	Pred micro concepts + UMLS terms	Text-only / Predicted Concepts	0.2179	0.0708	0.1889	2.5947	0.5716	0.2862	0.2130
Exp4	Pred coverage concepts + UMLS terms	Text-only / Predicted Concepts	0.2185	0.0718	0.1897	2.5289	0.5727	0.2876	0.2235
Exp5	Image-only	Image-only	0.2401	0.0806	0.2070	2.8757	0.6043	0.3385	0.3521
Exp6	Image + GT concepts	Image + GT Concepts	0.3656	0.1582	0.3205	6.6925	0.6656	0.4197	0.5850
Exp6	Image + Pred micro concepts	Image + Predicted Concepts	0.2504	0.0867	0.2159	3.5394	0.6019	0.3463	0.4088
Exp6	Image + Pred coverage concepts	Image + Predicted Concepts	0.2513	0.0870	0.2168	3.4938	0.6028	0.3460	0.4129

Appendix B: Supplementary Figures

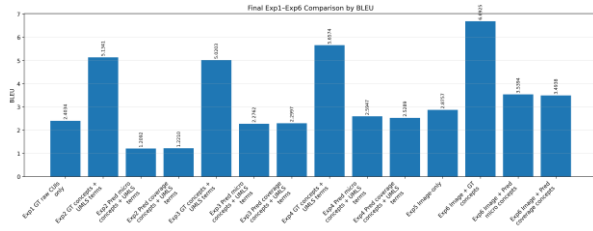


Figure B.1: Supplementary BLEU comparison across experiments.

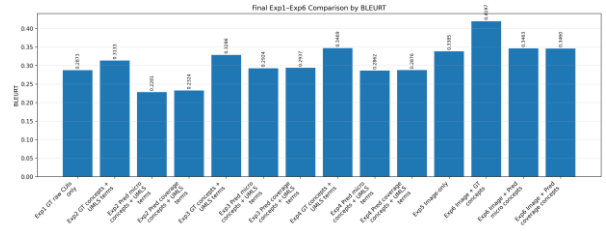


Figure B.2: Supplementary BLEURT comparison across experiments.

Figure B.1 confirms that the highest BLEU score is achieved by the Exp6 oracle setting. Figure B.2 confirms the strong semantic quality of the final oracle fusion model and the competitiveness of practical fusion settings.

Appendix C: Reproducibility Notes

All experiments were organized around fixed datasets, consistent validation subsets, saved checkpoints, and reusable CSV outputs. The main final tables include the Exp1-Exp6 metrics, best-result summaries, oracle-versus-practical gaps, reference comparisons, and error-breakdown outputs.

The complete code notebooks used during experimentation include the mixed GT/predicted T5 experiment, the image-only BiomedCLIP visual prefix baseline, and the final image-concept fusion experiment. Additional scripts generated final metric tables, figures, and qualitative error-analysis samples.

For final submission, the appendices are useful because they preserve supporting details without overloading the main chapters. If the department requires a shorter thesis, Appendix B can be reduced first because the main chapters already contain the essential figures.